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4
Olive Oil Composition

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Introduction

The composition of olive oil is primarily triacylglycerols (~99%) and secondarily free fatty acids, mono- and diacylglycerols, and an array of lipids such as hydrocarbons, sterols, aliphatic alcohols, tocopherols, and pigments. A plethora of phenolic and volatile compounds are also present. Some of these compounds contribute to the unique character of the oil.

Fatty Acids, Triacylglycerols, and Partial Glycerides

Fatty acids present in olive oil are palmitic (C16:0), palmitoleic (C16:1), stearic (C18:0), oleic (C18:1), linoleic (C18:2), and linolenic (C18:3). Myristic (C14:0), heptadecanoic and eicosanoic acids are found in trace amounts. Scano and co-workers (1999), using ^{13}C-Nuclear Magnetic Resonance Spectroscopy, detected traces of 11-*cis*-vaccenic and eicosenoic acids. Fatty acid compositional limits adopted in the most recent editions of Codex Alimentarius and International Olive Oil Council are given in Table 3.1.

IOOC also sets limits for *trans* fatty acids in each commercial category. For edible virgin olive oil categories the levels for C18:1*t* and for the sum of C18:2*t* and C18:3*t* isomers are extremely low (< 0.05% in each case).

Fatty acid composition may differ from sample to sample, depending on the zone of production, the latitude, the climate, the variety, and the stage of maturity of the fruit. Greek, Italian, and Spanish olive oils are low in linoleic and palmitic acids and they have a high percentage of oleic acid. Tunisian olive oils are high in linoleic and palmitic acids and lower in oleic acid. On the basis of the analysis of samples from various countries olive oils are classified in two types, one with a low linoleic-palmitic and high oleic acid content, and the other with a high linoleic-palmitic and low oleic acid content. Fatty acid composition of oil depends on the maturity stage. Ninni (1999) reported that oleic acid is formed first in the fruit and there is a strong

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TABLE 4.1
Fatty Acid Composition as Determined by Gas Chromatography (% m/m methyl esters)

Fatty Acid		Codex Alimentarius (2003)	IOOC*(2003)
lauric	C12:0	Not present in discernible amounts	Not specified
myristic	C14:0	< 0.1	< 0.05
palmitic	C16:0	7.5-20.0	7.5-20.0
palmitoleic	C16:1	0.3-3.5	0.3-3.5
heptadecanoic	C17:0	< 0.5	≤ 0.3
heptadecenoic	C17:1	< 0.6	≤ 0.3
stearic	C18:0	0.5-5.0	0.5-5.0
oleic	C18:1	55.0-83.0	55.0-83.0
linoleic	C18:2	3.5-21.0	3.5-21.0
linolenic	C18:3	**	≤ 1.0
arachidic	C20:0	0.8	≤ 0.6
eicosenoic	C20:1	Not specified	≤ 0.4
behenic	C22:0	< 0.3	≤ 0.2***
erucic	C22:1	Not present in discernible amounts	
lignoceric	C24:0	< 1.0	≤ 0.2

*The limits established include the precision values of the recommended method; **pending the results of IOOC survey and further consideration by the Committee on Fats and Oils, national limits may remain in place; ***Limit raised to <0.3 for olive-pomace oils.

antagonistic relationship between oleic and palmitic, palmitoleic and linoleic acids.

Characteristics and limits of physical and chemical indices and values, and of fatty acid composition for the various grades of virgin olive oils produced in each olive-growing area, determined at the onset and close to the olive oil production year are published yearly in national olive oil index files. Expansion of olive tree cultivation in countries of the Southern hemisphere (Australia, Argentina, New Zealand, and South Africa) as well as in California was the cause of a recent survey by the IOOC on the fatty acid composition of virgin olive oil produced around the world. Revision of Codex Alimentarius standards in order to be more representative of global production is therefore expected. Establishment of a maximum level for linolenic acid is of high priority as it can be used as a marker of adulteration. In the framework of this survey, 822 Australian olive oil samples were examined in the period 2002-03 at the Wagga Agricultural Institute (Mailer, 2005). Diversity in environment and cultivar characteristics had as a result wide ranges for the four major fatty acids (% methyl esters): (16:0), 7.8-18.8%; (18:1), 58.5-83.2%, (18:2), 2.8-21.1%; (18:3): 0.42-1.91%. The data deviated to a certain degree from those currently accepted by the Codex.

Slight deviations from the 1,3- random, 2-random distribution of fatty acids in the glycerol moiety of olive oil *triacylglycerols* have been found using advanced chromatographic and spectroscopic procedures (Santinelli et al., 1992; Vlahov, 2005). At the 2-position lower amounts of oleic and higher amounts of linoleic acid were evidenced in comparison to those obtained from calculations based on the theoretical

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random pattern. Deviations depend on the total amount of oleic or linoleic acid in the oil. Like other vegetable oils, olive oil has a high concentration of oleic acid and a low concentration of palmitic (≤ 2%) and stearic acids in position - 2 of the triacylglycerol molecules. The triacylglycerols found in significant proportions in olive oil are OOO (40-59%), POO (12-20%), OOL (12.5-20%), POL (5.5-7%) and SOO (3-7%) (Boskou, 1996). Smaller amounts of POP, POS, OLnL, LOL, OLnO, PLL, PLnO and LLL are also encountered (European Commission Regulation 282, 1998). Fully saturated moieties have not been reported and the same applies for the tri-unsaturated ones containing linolenic acid. Stearic and palmitic acids are absent from the 2-position of unsaturated species (tri- and tetraunsaturated) or from the molecule when there are more than five double bonds. Trilinolein or ECN 42 triacylglycerol content (as corrected recently), which is used as an authenticity marker by the EU, is the sum of the amounts of LLL, PoPoPo, SLnLn, PoPoL, PPoLn, OLLn, PLLn and PoOLn (positional isomers included). In the RP-HPLC chromatogram given in the same Regulation the triacylglycerol species assigned to the peaks according to elution order are LLL, OLnL, PLnL, LOL, OLnO, PLL, PLnO, LOL, PLO, OOO, SLO, POO, POL, SOO, SLS, and POS. Recent problems in the detection of hazelnut oil or almond oil in olive oil samples revitalized the interest in updating triacylglycerol compositional data (e.g. Parcerisa et al., 2000).

The presence of *partial glycerides* in olive oil is due either to incomplete triacylglycerol biosynthesis or hydrolytic reactions. In virgin olive oil, concentration of diacylglycerols (DG) range from 1 to 2.8% (Frega et al., 1993; Kiosseoglou and Kouzounas, 1993). In the diacylglycerol fraction C-34 and C-36 compounds prevail (Leone et al., 1988, Frega et al., 1993). Monoacylglycerols are present in much smaller quantities (less than 0.25%) whereas 1-species are considerably higher than the respective 2-monoglycerides. Their ratio depends on oil acidity (Paganuzzi, 1987; Paganuzzi, 1999). Storage conditions affect the distribution of fatty acids. 1,2-Diacylglycerols present in fresh oil tend to isomerize to the more stable 1,3-diacylglycerols. This rearrangement gives information about the age of the oil and storage conditions. The ratio of 1,3-/1,2-DG is considered as a useful criterion to monitor quality (Pérez-Camino et al., 2001; Spyros et al., 2004).

Hydrocarbons

Two hydrocarbons are present in considerable amounts in olive oil, squalene and β-carotene (the latter is discussed in the pigments section). Squalene (2,6,10,15,19,23-hexamethyl-2,6,10,14,18,22-tetracosahexaene) is the last metabolite preceding sterol ring formation. Its presence is regarded as partially responsible for the beneficial health effects of olive oil and its chemopreventive action against certain cancers (Rao et al., 1998, Smith et al., 1998). It is the major constituent of the unsaponifiable matter (see Glossary) and makes up more than 90% of the hydrocarbon fraction (Perrin, 1992; Lanzón et al., 1994). It ranges from 200 to 7500 mg per kg oil (Perrin, 1992).

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Even higher levels up to 12,000 mg/kg have been also reported (Lanzón et al., 1994). Squalene content depends on olive cultivar (De Leonardis et al., 1998; Manzi et al., 1998), oil extraction technology (Nergiz and Ünal, 1990), and it is dramatically reduced during the process of refining (Mariani et al., 1992; Lanzón et al., 1994). Variation in levels reported may be partially due to different analytical procedures used as commented by Nenadis and Tsimidou (2002).

Except for squalene, the hydrocarbon fraction of virgin olive oil is composed of diterpene and triterpene hydrocarbons, isoprenoidal polyolefins, and n-paraffins (Lanzon et al., 1994).

Tocopherols

Research on the occurrence and levels of tocopherols in virgin olive oils has shown that from the eight known "E-vitamers" the α-homologue comprises the 90% of the total tocopherol content. α-Tocopherol is found in the free form. The levels reported indicate a wide range of milligrams α-tocopherol per kg oil that depends on the cultivar potential and technological factors. Introduction of Good Manufacturing Practice and overall quality control programs supported by European Union for all edible types of olive oil had a positive impact on tocopherol levels in virgin olive oils of different origin. The levels currently found are much higher than the mean value of 100 mg/kg assigned to olive oil in the past (Gunstone et al., 1994; Belitz et al., 2004). This fact is important considering the *in vivo* antioxidant properties of the α-homologue (Kamal-Eldin and Appelqvist, 1996). Data for Italian and Spanish oils indicate a wide range for tocopherol levels: 55-264 mg/kg, n=18 (Conte et al., 1993); 97-315 mg/kg, n=52 (Fedeli and Cortesi, 1993); 100-320 mg/kg, n=23 (Esti et al., 1996); 147-187 mg/kg, n=6 (Ranalli and Angerosa, 1996); 103-283 mg/kg, n=14 (Cert et al., 1996);160-253 mg/kg, n=15 (Manzi et al., 1998); 55-234 mg/kg, n=65 (Salvador et al., 1998). Compared to them, Greek oils have α-tocopherol levels that are among the highest reported (e.g. 127-370 mg/kg for 1994-95; 98-333 mg/kg for 1995-96 and 100-365 mg/kg for 1996-97 crop seasons) (Psomiadou et al., 2000). Low amounts of the homologues β-tocopherol (~10 mg/kg), δ-tocopherol (~10 mg/kg) and γ-tocopherol (~20 mg/kg) are usually reported. The levels of α-tocopherol may be related to the high levels of chlorophyll pigments and the concomitant requirement for singlet oxygen deactivation (Grams and Eskins, 1972). Tocopherol concentration seems to be reduced in the ripe fruits. Data on the influence of the extraction system vary (Psomiadou and Tsimidou, 1998; Beltran et al., 2005). Refining or hydrogenation causes loss of tocopherols (Andrikopoulos et al., 1989; Rabascall and Riera, 1987).

Pigments

Virgin olive oil color is the result of green and yellow hues due to the presence of chlorophylls and carotenoids. It is influenced by olive cultivar, maturation index,

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production zone, extraction system, and storage conditions. Therefore it is considered as a quality index though no standardized method exists for its measurement.

Chlorophylls are encountered as pheophytins. Among the latter pheophytin α (Pheo α) is predominant (Mínguez-Mosquera et al., 1990; Mínguez-Mosquera et al., 1991; Rahmani and Csallany, 1991, Gandul-Rojas and Mínguez-Mosquera, 1996, Psomiadou and Tsimidou, 2001). The presence of Pheo α is related to processing conditions (Schwartz and Lorenzo, 1990), and enzymatic or enzymatic-like activity (Langmeier et al., 1993; Shioi et al., 1996). Handling and duration of storage cause further changes in pheophytin α content. The presence of pheophytin degradation products such as epimers, pyro-forms and allomers has been reported (e.g. Psomiadou and Tsimidou, 2001; Psomiadou and Tsimidou, 2002a). In a recent paper (Gallardo-Guerrero et al., 2005) the presence of pheophytin α degradation products was also evidenced for Spanish virgin olive oil samples stored under mild storage conditions for one year (15°C, dark, 3% headspace). These products, on the basis of previously reported findings, were identified as pyropheophytin α, 15^1-OH-lactone pheophytin α and 13^2-OH-pheophytin α. Under light exposure green pigments degrade causing oil bleaching (Psomiadou and Tsimidou, 2002b). Pheophytin α levels depend heavily on analytical methodology. Spectrometric methods result in higher levels than those determined with NP- or RP-HPLC. Chlorophyll α can only be found in recently obtained oils. Chlorophyll and pheophytin *b* are also present though in minute amounts.

The main carotenoids present in olive oil are lutein and β-carotene (Mínguez-Mosquera et al., 1990; Ranalli, 1992; Rahmani and Csallany, 1991; Gandul-Rojas and Mínguez-Mosquera, 1996). Levels reported are related to analytical method used. The presence of carotenoids in olive oil is closely related to that of green pigments and is influenced by the same factors. The carotenoid fraction may also include several xanthophylls (violaxanthin, neoxanthin, luteoxanthin, antheraxanthin, mutatoxanthin, and β-cryptoxanthin). The ratio between the two major carotenoids seems to be cultivar dependent (Gandul-Rojas and Mínguez-Mosquera, 1996; Psomiadou and Tsimidou, 2001).

Aliphatic and Aromatic Alcohols

Aliphatic and aromatic alcohols present in olive oil are found in free and esterified form. The most important are fatty alcohols and diterpene alcohols. Alkanols and alkenols with less than ten carbon atoms in their molecule, which are present in free and esterified form, and some aromatic alcohols (benzyl alcohol and 2-phenylethanol) are constituents of the olive oil volatile fraction. Benzyl esters of hexacosanoic and octacosanoic acid have been also found in olive oil (Reiter and Lorbeer, 2001).

Fatty Alcohols. This class of minor constituents consists of linear saturated alcohols with more than 16 carbon atoms which are present in the free and esterified form.

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The main fatty alcohols present in olive oil are docosanol, tetracosanol, hexacosanol, and octacosanol (Tiscornia et al., 1982; Boskou et al., 1983; Frega et al., 1992). Fatty alcohols with odd carbon atoms (tricosanol, pentacosanol, and heptacosanol) may be found in trace amounts (Tiscornia et al., 1982; Boskou et al., 1983).

Virgin olive oil total fatty alcohol levels are not usually higher than 250 mg/kg (Grob et al., 1990; Ranalli and Angerosa, 1996; Ranalli and Serraiocco, 1996; Cert et al., 1999; Aparicio and Luna, 2002). Tetracosanol and hexacosanol were found to be present at higher levels than the other fatty alcohols (Boskou et al., 1983; Ranalli and Angerosa, 1996; Ranalli and Serraiocco, 1996; Aparicio and Luna, 2002). Much higher levels were found in solvent extracted olive oils (Paganuzzi and Leoni, 1979; Tiscornia et al., 1982; Boskou et al., 1983; Grob et al., 1990). However, refined solvent extracted olive oils were found to contain fatty alcohols at concentrations similar to that of virgin olive oils (Grob et al., 1990). Fatty alcohol content is affected by cultivar, crop year, fruit ripeness, and processing (Camera and Angerosa, 1978; Ranalli and Angerosa, 1996; Ranalli and Serraiocco, 1996; Cert et al., 1999; Aparicio and Luna, 2002).

Esters of fatty alcohols with fatty acids (waxes) are important minor olive oil constituents because they can be used as a criterion to differentiate various olive oil types (EC Regulation 2568, 1991). The main waxes detected in olive oil are esters of oleic or palmitic acid with 36, 38, 40, 42, 44, and 46 carbon atoms (Reiter and Lorbeer, 2001). Virgin olive oils contain waxes at levels lower than 150 mg/kg, while raw and refined solvent extracted olive oils have wax content higher than 2000 mg/kg (Grob et al., 1990). This difference is officially used for the distinction between olive oil and solvent extracted olive oil. Wax content and composition are affected by cultivar, crop year, and processing (Ranalli and Angerosa, 1996; Ranalli and Serraiocco, 1996; Cert et al., 1999).

Diterpene Alcohols. Phytol and geranylgeraniol are two acyclic diterpenoids (Fig.4.1) present in the aliphatic alcohol fraction of olive oil in the free and esterified form (Camera and Angerosa, 1978; Paganuzzi and Leoni, 1979; Mariani et al., 1992; Cert et al., 1999; Reiter and Lorbeer, 2001). Phytol, which probably originates from chlorophyll, has been found in monovarietal virgin olive oils at levels ranging from 25 to 595 mg/kg (Aparicio and Luna, 2002). Geranylgeraniol is reported to be present in virgin olive oil from a new olive cultivar – I-77 at levels lower than 50 mg/kg (Ranalli et al., 2000). Its levels are used in the calculation of the alcoholic index, a useful parameter for detecting solvent extracted olive oil in virgin olive oil. Esters identified in the wax fraction of extra virgin olive oil are oleate, eicosanoate, eicosenoate, docosanoate, and tetracosanoate (Reiter and Lorbeer, 2001). They are mainly phytyl derivatives.

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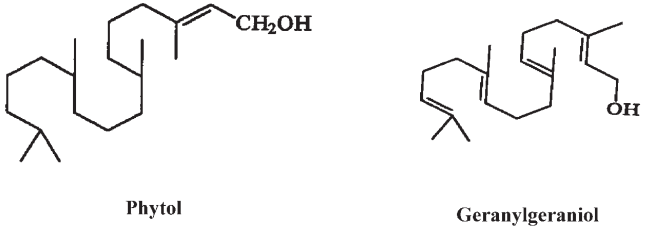


Fig. 4.1. Structural formulae of the olive oil diterpene alcohols.

Sterols

Sterols are important lipids related to the quality of the oil and broadly used for checking its genuineness. Four classes of sterols occur in olive oil: common sterols (4-desmethylsterols), 4α-methylsterols, triterpene alcohols (4, 4-dimethylsterols), and triterpene dialcohols.

Common Sterols (4α-desmethylsterols). Olive oil contains common sterols mainly in free and esterified form (Grob et al., 1990). However, these sterols have also been found as sterylglucosides and lipoproteins (Homberg and Bielefeld, 1985). The main components of this sterol fraction are β-sitosterol, Δ5-avenasterol, and campesterol (Itoh et al., 1973a; Boskou and Morton, 1975; Kornfeldt, 1981). Other sterols present in smaller quantities or in trace amounts are stigmasterol, cholesterol, brassicasterol, chlerosterol, ergosterol, sitostanol, campestanol, Δ7-avenasterol, Δ7-cholestenol, Δ7-campestenol, Δ7-stigmastenol, Δ5,23-stigmastadienol, Δ5,24-stigmastadienol, Δ7,22-ergostadienol, Δ7,24-ergostadienol, 24-methylene-cholesterol, and 22,23-dihydrobrassicasterol (Itoh et al., 1981; Calapaj et al., 1993; Mariani et al., 1995; Mariani, 1998). Structural formula of sterols reported to be present in olive oil are given in Figure 4.2.

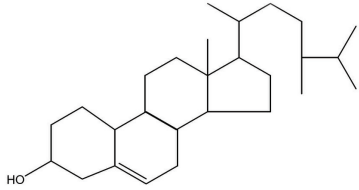
Total sterol content of virgin olive oils varies mainly between 1000 mg/kg, which is the lower limit set by the European Union Commission (EC Regulation 2568, 1991), and 2000 mg/kg (Morchio et al., 1987; Aparicio and Luna, 2002). Lampante olive oils contain higher amounts of total sterols (Morchio et al., 1987; Grob et al., 1990). Refined olive oils contain total sterols at lower levels because the refining process gives rise to significant losses of sterols, which may be as high as 25% (Morchio et al., 1987). Total sterol content of solvent extracted olive oils is up to three times higher than that of virgin olive oils (Morchio et al., 1987).

Studies on olive oil sterol composition show that β-sitosterol makes up 75 to 90% of the total sterol fraction, while Δ^5-avenasterol usually ranges between 5% and 20% (Itoh et al., 1981; Calapaj et al., 1993). Percentages of Δ^5-avenasterol, up to 36%, have been reported for Greek virgin olive oils (State Chemical Laboratory, 1994).

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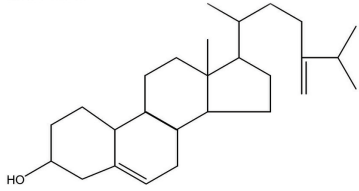
Cholesterol

Δ^5-Cholesten-3β-ol



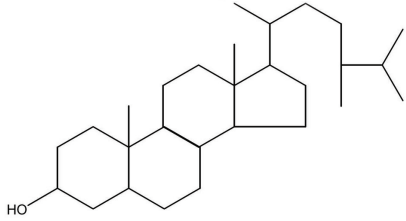
24-Methylene-cholesterol

24-Methylene-Δ^5-cholesten-3β-ol



Campestanol

(24R)-24-Methyl-cholestan-3β-ol



Campesterol

(24R)-24-Methyl-Δ^5-cholesten-3β-ol

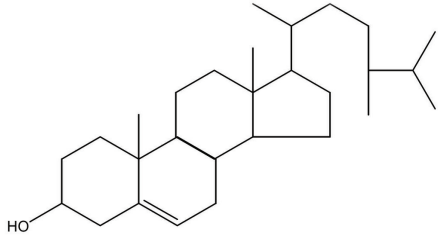
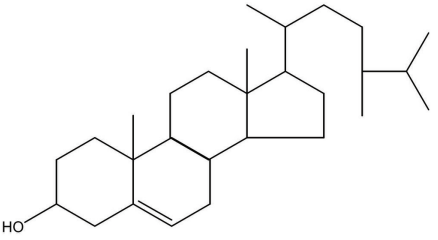


Fig. 4.2. 4α-Desmethylsterols present in olive oil.

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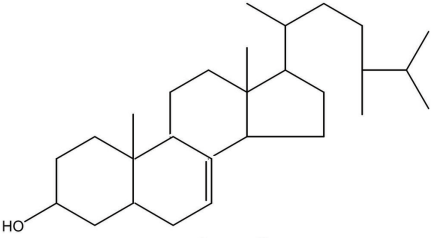
22,23-Dihydrobrassicasterol

(24S)-24-Methyl-Δ^5-cholesten-3β-ol



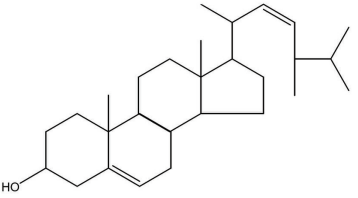
Δ^7-Campestenol

(24R)-24-Methyl-Δ^7-cholesten-3β-ol



Ergosterol

(24R)-24 Methyl-$\Delta^{5,22}$-cholestadien-3β-ol



Brassicasterol

(24S)-24 Methyl-$\Delta^{5,22}$-cholestadien-3β-ol

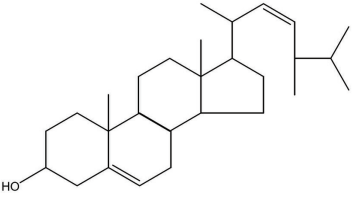
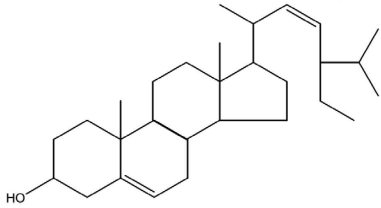


Fig. 4.2. Continued

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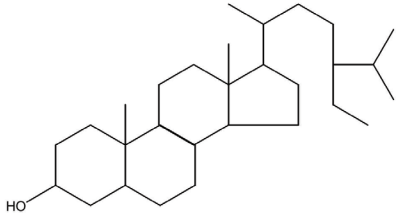
Stigmasterol

(24R)-24 Ethyl-$\Delta^{5,22}$-cholestadien-3β-ol



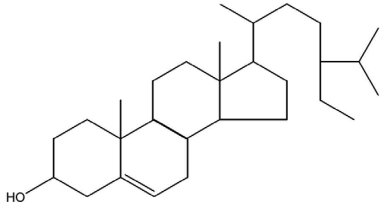
Sitostanol

(24R)-24-Ethyl-cholestan-3β-ol



β-Sitosterol

(24R)-24-Ethyl-Δ^{5}-cholesten-3β-ol



Δ^{7}-Stigmastenol

(24R,S)-24-Ethyl-Δ^{7}-cholesten-3β-ol

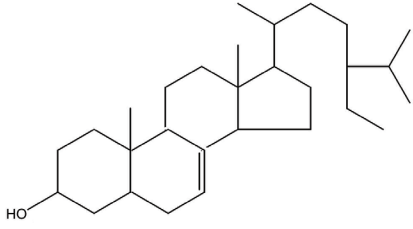
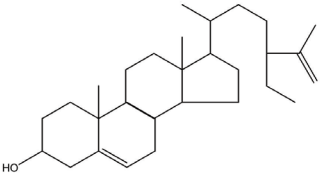


Fig. 4.2. Continued

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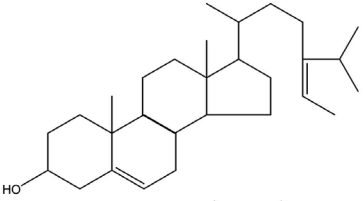
Chlerosterol

(24S)-24-Ethyl-$\Delta^{5,25}$-cholestadien-3β-ol



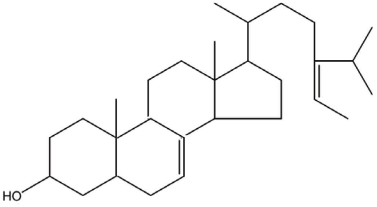
Δ^{5}-Avenasterol

(24Z)-24-Ethylidene-Δ^{5}-cholesten-3β-ol



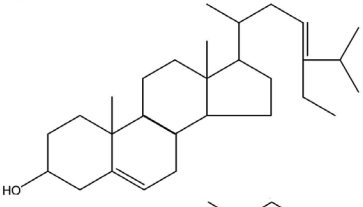
Δ^{7}-Avenasterol

(24Z)-24-Ethylidene-Δ^{7}-cholesten-3β-ol



$\Delta^{5,23}$-Stigmastadienol

(24R,S)-24-Ethyl-$\Delta^{5,23}$-cholestadien-3β-ol



$\Delta^{5,24}$-Stigmastadienol

(24R,S)-24-Ethyl-$\Delta^{5,24}$-cholestadien-3β-ol

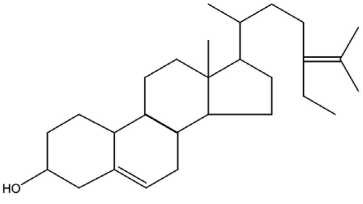
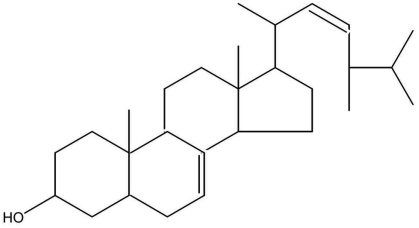


Fig. 4.2. Continued

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$\Delta^{7,22}$-Ergostadienol

(24R)-24 Methyl-$\Delta^{7,22}$-cholestadien-3β-ol



$\Delta^{7,24}$-Ergostadienol

(24R)-24 Methyl-$\Delta^{7,24}$-cholestadien-3β-ol

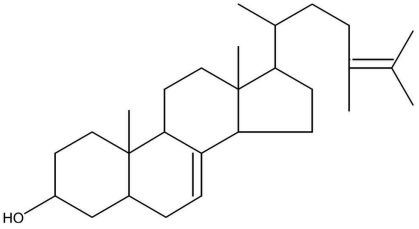


Fig. 4.2. Continued

From the other sterols present in olive oil, campesterol and stigmasterol make up 4% and 2% of the total sterol fraction, respectively (Calapaj et al., 1993), but higher values have also been determined (Paganuzzi, 1985; Lanuzza et al., 1995; Rivera del Alamo et al., 2004). In all cases, the percentage of campesterol is higher than that of stigmasterol. The rest of the sterols occur in minute quantities. The levels of Δ^5- and Δ^7-avenasterol, Δ^7-stigmastenol, stigmasterol, and chlerosterol differentiate virgin, refined, and solvent extracted olive oils (De Blas and del Valle González, 1996).

Approximately 10 to 40% of total sterols are present as steryl esters (Boskou and Vlachopoulou, 1986; Grob et al., 1990). According to Mariani and his co-workers (1991), the composition of the two sterol fractions differs. Diunsaturated Δ^5-sterols (Δ^5-avenasterol, stigmasterol, and brassicasterol) are present at relatively higher levels in the free than in the esterified form. The opposite is the case of diunsaturated Δ^7-sterols. Raw and refined lampante and solvent extracted olive oils have been found to contain sitosteryl-C_{18}-esters at significantly higher levels than extra virgin olive oils (Grob et al., 1990). Therefore, the percentage of free β-sitosterol in total β-sitosterol can be used as a key parameter for assessing the quality and genuineness of a virgin olive oil. In the case of solvent extracted olive oils, losses during refining are more pronounced in the free sterol fraction (Grob et al., 1990).

Sterol composition and total sterol content are affected by cultivar, crop year, degree of fruit ripeness, storage time of fruits prior to oil extraction, processing, and also by geographic factors (Fedeli, 1993; Ranalli and Angerosa, 1996; Ranalli and

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Serraiocco, 1996; Ranalli et al., 1997; Salvador et al., 1998; Cert et al., 1999; Gutiérrez et al., 1999; Koutsaftakis et al., 1999; Ranalli et al., 1999; Gutiérrez et al., 2000, Ranalli et al., 2000, Aparicio and Luna, 2002; Rivera del Alamo et al., 2004). Olive storage was found to be responsible for notable changes in the levels of individual sterols (Camera et al., 1978). Olive harvesting practices also affect individual sterol levels. Fedeli (1993) claims that stigmasterol rises to higher levels in olives left on the ground. Sterol composition is also affected by the refining process. Mariani and his coworkers (1992) studied the qualitative and quantitative variations occurring in both the free and esterified sterol fraction during bleaching with activated earths. They found that the esterified sterol fraction is subjected to more limited reductions than the free sterol fraction which is sometimes completely lost.

4-Methylsterols. Olive oil contains 4-monomethylsterols in small quantities (Fig. 4.3.), which are intermediates in sterol biosynthesis. They are present in free and esterified form (Kiosseoglou et al., 1987; Chryssafidis et al., 1992). The predominating components are obtusifoliol, gramisterol, cycloeucalenol, and citrostadienol (Itoh et al., 1973b; Boskou and Morton, 1975; Paganuzzi and Leoni, 1979; Itoh et al., 1981). They are Δ7- or Δ8-sterols except cycloeucalenol which has a 9,19-cyclopropane ring in the steroid skeleton.

The 4α-methylsterol fraction is extremely complex and gas chromatographic analysis reveals a considerable number of unknown minor components. Itoh and his co-investigators (1981) identified among these minor components 24-methyl-31-nor-9(11)-lanosterol, 24-methylene-31-nor-9(11)-lanosterol, 24-ethyllophenol, 24-methyl-(E)-23-dehydrolophenol, 24-ethyl-(E)-23-dehydrolophenol, 24-methyl-31-nor-(E)-23-dehydrocycloartanol, 28-isocitrostadienol, 24-ethyl-24(25)-dehydrolophenol, and 24-methyl-24(25)-dehydrolophenol.

The levels of total 4α-methylsterols are lower than that of common sterols and triterpene alcohols and vary between 50 and 360 mg/kg (Itoh et al., 1973b; Cert at al, 1999; Aparicio and Luna, 2002). In solvent extracted olive oils these levels are higher (Paganuzzi and Leoni, 1979; Kornfeldt, 1981). Citrostadienol has been found in oils obtained from fruits of a new olive cultivar – I-77 at levels up to 100 mg/kg (Ranalli et al., 2000). 4α-methylsterol content is affected by olive cultivar (Aparicio and Luna, 2002).

Triterpene Alcohols (4,4-dimethylsterols). The main components of the very complex 4,4-dimethylsterol fraction are β-amyrin, butyrospermol, 24-methylenecycloartanol, and cycloartenol (Fig. 4.4.) (Itoh et al., 1973b). Other triterpene alcohols identified in olive oil, but present in smaller quantities or in trace amounts, are cyclobranol, cyclosadol, dammaradienol, germanicol, 24-methylene-24-dihydroparkeol, taraxerol, α-amyrin, 7, 24-tirucalladienol, parkeol, and tirucallol (Itoh et al., 1981). Triterpene alcohols are present in the free and esterified form (Mariani et al., 1998). Significant

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differences were observed between the distribution patterns of the total and esterified triterpene alcohol fraction in virgin olive oils, especially in the content of 24-methylenecycloartanol, butyrospermol, and cycloartenol (Chryssafidis et al., 1992). The triterpene alcohol fraction of olive oil and solvent extracted olive oil differ also in composition (Itoh et al., 1981; Grob et al., 1990).

Total triterpene alcohol levels have been found to vary between 350 and 1500 mg/kg (Kiosseoglou et al., 1987; Ranalli and Angerosa, 1996; Ranalli and Serraioco, 1996; Ranalli et al., 1997; Cert at al, 1999; Ranalli et al., 1999; Aparicio and Luna, 2002). Grob and his co-investigators (1990) determined the levels of free cycloartenol in virgin olive oils, lampante olive oils and solvent extracted olive oils. Italian virgin olive oils were found to contain cycloartenol at higher levels than Greek or Spanish virgin olive oils and Italian solvent extracted olive oils.

Triterpene alcohol composition and total triterpene alcohol content is affected by cultivar, crop year, and processing (Grob et al., 1990; Nergiz and Ünal K, 1990; Ranalli and Angerosa, 1996; Ranalli and Serraioco, 1996; Ranalli et al., 1997; Cert at al, 1999; Ranalli et al., 1999; Aparicio and Luna, 2002).

Significant structural modifications take place in the triterpene alcohol fraction during refining. Strocchi and Savino (1989) claim that when the oil is bleached with activated clays the 9,19-cyclopropane ring of 24-methylenecycloartanol opens giving rise to the formation of three derivatives (isomers which have a double bond at the carbon atom C_7, C_8, or C_9). A Δ^7-isomer of 24-methylenecycloartanol (24-methyl-5α-lanosta-7,24-dien-3β-ol) has also been identified in refined olive oils (Lanzón et

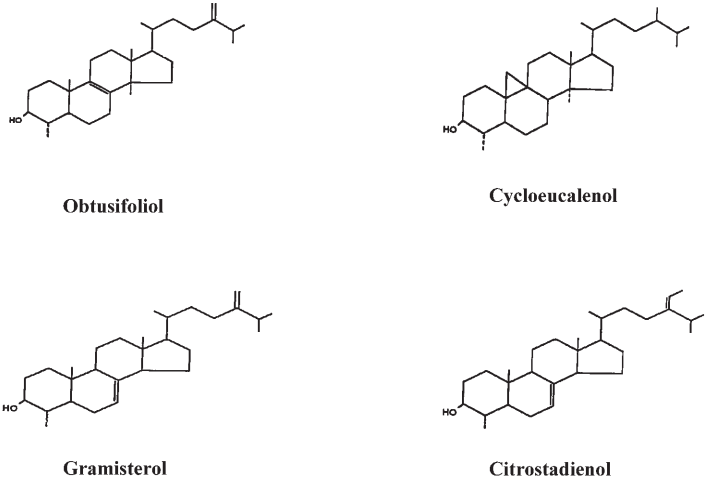


Fig. 4.3. Structural formulae of the main olive oil 4α-methylsterols.

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al., 1999). Qualitative and quantitative variations occurring in olive oil free and esterified triterpene alcohol fractions during bleaching with activated earths were evaluated by Mariani and his co-investigators (1992). They found a remarkable reduction of free triterpene alcohols and a more limited reduction of esterified triterpene alcohols.

Triterpene Dialcohols. Erythrodiol (homo-olestranol, 5α-olean-12-ene-3β, 28-diol) and uvaol (Δ12-ursen-3β,28-diol) are the main triterpene dialcohols identified in olive oil (Fig. 4.5.). Erythrodiol was found to be present in the free and esterified form (Mariani et al., 1998). Its content in olive oil is mainly affected by cultivar (Aparicio and Luna, 2002).

Virgin olive oils were found to contain total erythrodiol at levels ranging from 19 to 69 mg/kg (Aparicio and Luna, 2002) and free erythrodiol at levels usually lower than 50 mg/kg (Grob et al., 1990). Total and free erythrodiol levels of solvent extracted olive oils were found to be much higher than that of olive oils (Mariani et al., 1987; Grob et al., 1990).

The sum of erythrodiol and uvaol levels is usually given as a percentage of the whole sterol fraction because triterpene dialcohols are co-chromatographed with 4α-desmethylsterols. This sum, which must not exceed a limit set by the European Union Commission (Regulation 1989/2003), is used as a reliable indicator for distinguishing olive oil and solvent extracted olive oil (see Chapter 7, Analysis and Authentication).

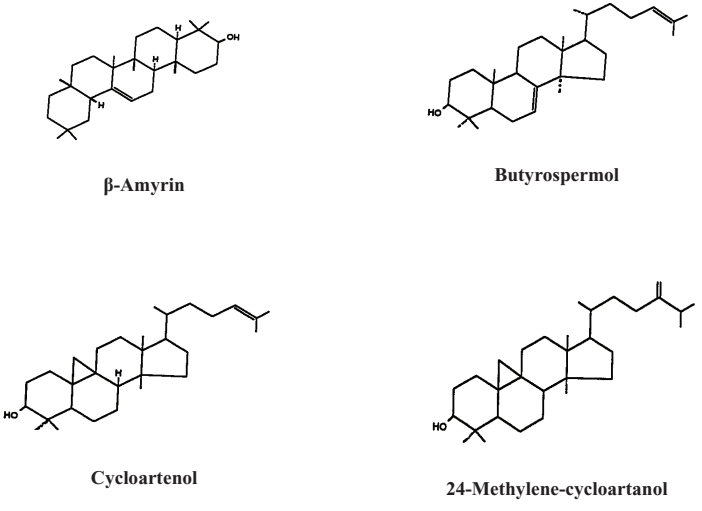


Fig. 4.4. Structural formulae of the main olive oil triterpene alcohols.

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TABLE 4.2
Composition of Total and Esterified Triterpene Alcohol Fraction of Six Virgin Olive Oils

		Origin					
		Halkidiki	Kavala	Fthiotida	Keffalinia	Lesvos	Crete
	RRT/Chol	Total/ Esterified	Total/ Esterified	Total/ Esterified	Total/ Esterified	Total/ Esterified	Total/ Esterified
Cholesterol	1,00						
	1,13	Trace, 1	1, –	1, 5	–, –	–, –	4, –
β-Amyrin	1,16	2, 4	2, –	1, 6	1, –	2, 3	7, 7
Butyrospermol	1,17	7, 21	12, 17	7, 14	9, 27	4, 14	22, 37
	1,20	–, –	–, 4	–, –	–, –	–, –	–, –
Cycloartenol	1,21	1, 6	18, 51	12, 45	41, 50	21, 39	32, 46
	1,23	–, 2	–, –	–, 6	–, 3	–, –	–, –
24-Methylene cycloartanol	1,28	64, 31	63, 22	77, 22	47, 14	62, 44	28, 6
	1,29	–, –	1, –	–, 1	–, –	–, –	2, 2
	1,31	2, Trace	1, –	–, –	–, –	4, Trace	2, 1
	1,33	–, –	2, –	–, 1	–, –	7, –	3, 1
	1,35	–, 1	Trace, 1	2, –	2, –	–, –	–, Trace

Source: Chryssafidis et al. (1992). Reprinted with permission of J. Sci. Food Agric.

The relative triterpene dialcohol content in the whole sterol fraction is affected by cultivar and processing (Ranalli and Angerosa, 1996; Ranalli and Serraiocco, 1996; Ranalli et al., 1997; Cert at al, 1999; Koutsaftakis et al., 1999; Ranalli et al., 1999; Ranalli et al., 2000).

Triterpene Acids

Hydroxy pentacyclic triterpene acids are important olive fruit constituents (Fig. 4.6.). They are biologically active compounds and are present at trace amounts in olive oil. Oleanolic (3β-hydroxyolean-12-en-28-oic acid) and maslinic acid (2α, 3β-dihydroxyolean-12-en-28-oic acid) are the main triterpene acids present in virgin olive oil because they occur in the olive husk and a small quantity may be extracted during processing (Caputo et al., 1974). Both compounds and traces of ursolic acid (3β-hydroxyurs-12-en-28-oic acid) are also located in the reticular lipid layer of olive skin (Bianchi et al., 1994). Betulinic acid (3β-hydroxylup-20-(29)-en-28-oic acid) has also been identified in the skin of Coratina olive cultivar (Bianchi et al., 1992). Esterified derivatives of the triterpene acids were not found in olives and olive oils (Pérez-Camino and Cert, 1999).

Pérez-Camino and Cert (1999) claim that the main factor contributing to a higher level of hydroxy pentacyclic triterpene acids is oil acidity. Olive cultivar, olive ripeness, and oil extraction system have less influence on the levels of these acids. Total triterpene acid content of extra virgin olive oils obtained from fruits of different

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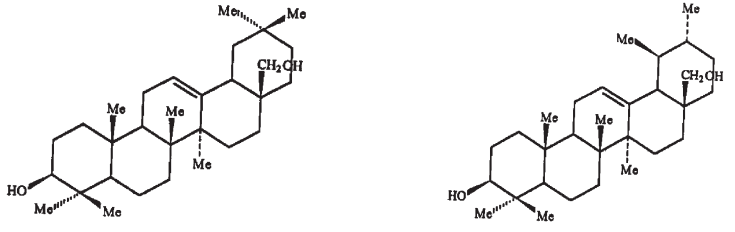


Fig. 4.5. Structural formulae of the olive oil triterpene dialcohols.

olive cultivars was found to range between 40 and 185 mg/kg. Virgin olive oils with free acidity higher than 1.0% and solvent extracted olive oils were found to contain these acids at levels higher than 300 mg/kg and 2400 mg/kg, respectively. The oleanolic and maslinic acid levels of virgin olive oils were found to be similar. Ursolic acid was present in traces. Oleanolic acid content of solvent extracted olive oils was much higher than that of maslinic acid. During chemical refining, total losses of oleanolic and maslinic acid were observed. Significant losses were also observed during physical refining (50-80% for oleanolic and 60-80% for maslinic acid).

Volatile and Aroma Compounds

Volatile compounds present in olive oil have been reported by Flath et al. (1973), Olías et al. (1977), Montedoro et al. (1978), Olías et al. (1980), Gutiérrez et al. (1981), Guth and Grosch (1991), Morales et al. (1994), Angerosa et al. (1998), Reiners and Grosch (1998), Bortolomeazzi et al. (2001), Cavalli et al. (2003), Servili et al. (2003), Vichi et al. (2003), Cavalli et al. (2004), and also Morales et al. (2005). Approximately two hundred and eighty compounds have been identified in the volatile fraction of virgin olive oils. They are hydrocarbons (more than 80 compounds), alcohols (45 compounds), aldehydes (44 compounds), ketones (26 compounds), acids (13 compounds), esters (55 compounds), ethers (5 compounds), furan derivatives (5 compounds), thiophene derivatives (5 compounds), pyranones (1 compound), thiols (1 compound), and pyrazines (1 compound). From this large number of compounds, only 67 were found to be present at levels higher than their odor threshold contribute to the aroma of virgin olive oil (see table 4.3.). About twenty of these compounds contribute to the flavor of virgin olive oils with sensory defects (Sanchez Saez et al., 1991; Angerosa et al., 1996; Morales et al., 2005).

The potent odorants of olive oil have been evaluated by applying aroma extract dilution analysis (AEDA) and gas chromatography-olfactometry analysis of headspace samples (Guth and Grosch, 1991; Blekas et al., 1994; Blekas and Guth, 1995; Rein-

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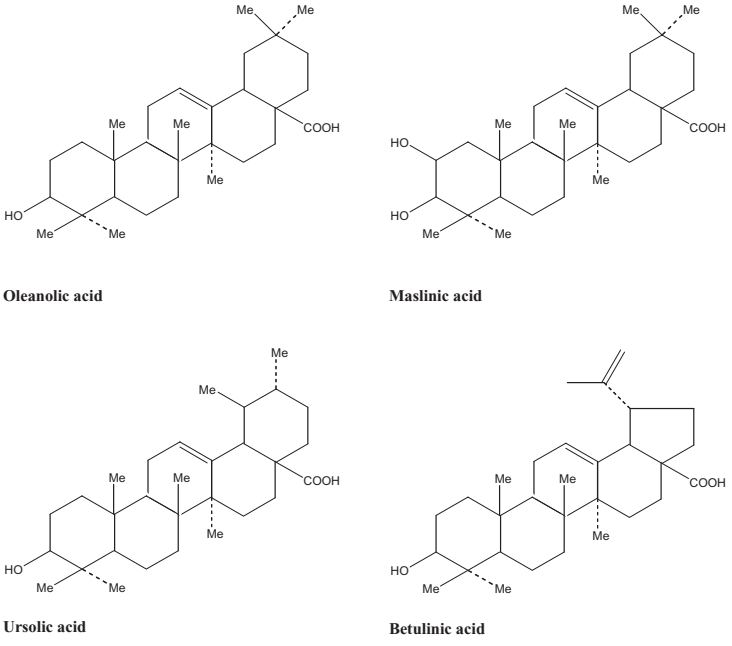


Fig. 4.6. Structural formulae of the olive oil hydroxy pentacyclic triterpene acids.

ers and Grosch, 1998). AEDA is a screening method which is applied to the volatiles isolated by a high vacuum codistillation of oil with diethyl ether. An aliquot of the concentrated by distillation and microdistillation sample is analyzed by HRGC while the effluent of the capillary column is sniffed. The aliquot is then diluted with diethyl ether in volume ratios 1 to 2^n-1 (n = 1, 2, 3, etc) and the new samples are analyzed by gas chromatography-olfactometry until no odor is detected. In this way the flavor dilution factor (FD-factor) of each odorous compound can be estimated. The FD-factor is the highest dilution at which a volatile compound is still smelled during gas chromatography-olfactometry analysis and it is defined as the ratio of concentration of the odorant in the initial extract to its concentration in the most dilute extract in which odor is detected. FD-factor is proportional to the Odor Activity Value (OAV), which is the ratio of concentration to odor threshold of the compound in odorless oil (Grosch, 1994). The OAVs show the actual contribution of each odorant to the olive oil aroma (Guth and Grosch, 1993; Blekas et al., 1994; Blekas and Guth, 1995; Reiners and Grosch, 1998; Morales et al., 2005). The odor threshold values of olive oil odorants, which are determined retronasally (by tasting), are given in Table 4.4.

Table 4.5 presents odorants identified in Greek olive oils. Odor description des-

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TABLE 4.3
Odorants Contributing to the Aroma of Olive Oil

Aldehydes	Esters	Alcohols
Ethanal	Ethyl acetate	Butan-2-ol
Propanal	Ethyl propanoate	2-Methylbutan-1-ol
2-Methylbutanal	Ethyl butanoate	3-Methylbutan-1-ol
3-Methylbutanal	Ethyl octanoate	(Z)-3-Hexen-1-ol
Pentanal	Ethyl cinnamate	Heptan-2-ol
Hexanal	Ethyl 2-methylpropanoate	1-Octen-3-ol
(E)-2-Hexenal	Ethyl 2-methylbutanoate	Nonan-1-ol
(Z)-3-Hexenal	Ethyl 3-methylbutanoate	2-Phenylethanol
Heptanal	Ethyl cyclohexylcarboxylate	
(E)-2-Heptenal	Butyl acetate	**Ketones**
Octanal	3-Methylbutyl acetate	1-Penten-3-one
(E)-2-Octenal	(Z)-3-Hexenyl acetate	Octan-2-one
Nonanal	2-Methylpropyl butanoate	1-Octen-3-one
(E)-2-Nonenal		2-Methyl-2-hepten-2-one
(Z)-2-Nonenal	**Acids**	(Z)-1,5-Octadien-3-one
(Z)-3-Nonenal	Acetic acid	(E)-β-Damascenone
(E,E)-2,4-Nonadienal	Propanoic acid	(Z)-β-Damascenone
(E,Z)-2,6-Nonadienal	Butanoic acid	
(E)-2-Decenal	Pentanoic acid	**Others**
(Z)-2-Decenal	2-Methylbutanoic acid	Guaiacol
(E,E)-2,4-Decadienal	3-Methylbutanoic acid	4-Ethylguaiacol
(E,Z)- 2,4-Decadienal	Hexanoic acid	1-Octen-3-hydroperoxide
trans-4,5-Epoxy-(E)-2-decenal	Heptanoic acid	4-Methoxy-2-methyl-2-butanethiol
Phenylacetaldehyde		2-Isobutyl-3-methoxypyrazin
Vanillin		n-Octane

ignated during AEDA and FD-factors on an SE-54 capillary column are also given in the table (Blekas and Guth, 1995; Blekas and Guth, unpublished data).

By applying gas chromatography-olfactometry analysis of headspace samples to the screening for potent odorants, Blekas et al. (1994), and also Reiners and Grosch (1998), found also that ethanol also contributes to virgin olive oil aroma.

Levels of virgin olive oil odorants in commercially available samples with different aroma characteristics originated from Greece (a, b, c, d), Italy (e, f), and Spain (g, h) are given in Table 3.6. The odorants were quantified by stable isotope dilution analysis (Blekas and Guth, unpublished data).

It is clear that the qualitative composition of odorants present in virgin olive oils is similar. However, the quantitative composition differs strongly when the aroma characteristics are different. Olive oil obtained from healthy olives, harvested at the right ripening stage, and by proper extraction techniques contains mainly volatiles derived from linoleic and linolenic acid decomposition through the lipoxygenase pathway (Olias et al., 1993). The most abundant compounds are hexanal, (E)-2-hexenal, (Z)-3-hexenal, hexan-1-ol, (Z)-3-hexen-1-ol, hexyl acetate, and (Z)-3-hex-

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enyl acetate. These volatiles are responsible for the green and fruity perception of the unique virgin olive oil aroma. C_5 aldehydes, ketones and alcohols, and pentene dimers also arise from a cleavage reaction of 13-hydroperoxide of linolenic acid (Salch et al., 1995), are usually present at levels lower than their odor threshold. The formation of these C_6 and C_5 compounds is affected by the cultivar, the degree of fruit ripeness, the storage time of fruits prior to oil extraction, and by the processing (Blekas et al., 1994; Aparicio and Morales, 1998; Angerosa et al., 1998a; Lercker et al., 1999; Morales et al., 1999; Salas and Sánchez, 1999; Angerosa et al., 1999; Morales and Aparicio, 1999; Angerosa et al., 1999a; Angerosa et al., 2000; Angerosa et al., 2001; Aparicio and Luna, 2002; Ridolfi et al., 2002; Servili et al., 2002; Angerosa and Basti, 2003; Benincasa et al., 2003; Luaces et al., 2003; Pérez et al., 2003; Servili et al., 2003; Tura

TABLE 4.4
Odor threshold of olive oil odorants determined retronasally

Compound	Odor threshold (μg/kg)	Compound	Odor threshold (μg/kg)
Ethanal	7,1[d]	(E)-β-Damascenone	3,7[c]
2-Methylbutanal	8,2[c]	2-Methylbutan-1-ol	480[e]
3-Methylbutanal	10,8[b]	3-Methylbutan-1-ol	100[e]
Pentanal	150[d]	(Z)-3-Hexen-1-ol	360[c]
Hexanal	75[d]	2-Phenylethanol	120[c]
(E)-2-Hexenal	250[d]	Ethyl acetate	940[e]
(Z)-3-Hexenal	1,2[d]	Ethyl propanoate	100[e]
Heptanal	50[d]	Ethyl 2-methylpropanoate	0,75[a]
(E)-2-Heptenal	400[d]	Ethyl butanoate	3,5[c]
Octanal	55[d]	Ethyl 2-methylbutanoate	0,75[a]
(E)-2-Octenal	125[d]	Ethyl 3-methylbutanoate	0,50[c]
Nonanal	260[d]	Ethyl cyclohexylcarboxylate	0,06[c]
(E)-2-Nonenal	65[d]	Butyl acetate	300[e]
(Z)-2-Nonenal	0,6[d]	(Z)-3-Hexenyl acetate	750[a]
(Z)-3-Nonenal	35[d]	2-Methylpropyl butanoate	100[e]
(E,E)-2,4-Nonadienal	460[d]	Guaiacol	13[c]
(E,Z)-2,6-Nonadienal	1,5[d]	Acetic acid	380[c]
(E)-2-Decenal	150[d]	Propanoic acid	720[e]
(E,E)-2,4-Decadienal	40[d]	Butanoic acid	650[e]
6-Methyl-5-hepten-2-one	1000[e]	3-Methylbutanoic acid	25[c]
1-Penten-3-one	3,2[d]	Pentanoic acid	600[e]
1-Octen-3-one	0,3[d]	Hexanoic acid	700[e]
(Z)-1,5-Octadien-3-one	0,03[d]	Heptanoic acid	100[e]
trans-4,5-Epoxy-(E)-2-decenal	3[d]	4-Methoxy-2-methyl-2-butanethiol	0,025[c]

[a] Guth and Grosch, 1993
[b] Preininger and Grosch, 1994
[c] Reiners and Grosch, 1998
[d] Belitz et al., 2004
[e] Morales et al., 2005

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TABLE 4.5
Odorants which contribute to the aroma of Greek virgin olive oils with different flavor profiles

Compound	RI on capillary SE-54	RI on capillary OV-1701	Odor description (assigned during AEDA)	FD-factor* on capillary SE-54 Samples A	B	C	D
3-Methylbutanal	650	731	Malty	2^3	2^3	2^2	2^3
1-Penten-3-one	680	775	Sharp, pungent	2	2	2^4	2^4
3-Methylbutan-1-ol	735	848	Malty	2	1	<1	1
Ethyl 2-methylpropanoate	765	817	Fruity	2^4	2^5	2^4	2^6
(Z)-3-Hexenal	797	890	Green, apple-like	2^5	2^6	2^{11}	2^9
Hexanal	800	883	Green, grassy	2^8	2^8	2^6	2^4
Ethyl 2-methylbutanoate	849	906	Fruity	2^6	2^7	2^6	2^8
(E)-2-Hexenal	855	964	Green, bitter	2^3	2^3	2^5	2^6
(Z)-3-Hexen-1-ol	859	973	Green leaves	2^3	2	2^3	2^2
Heptanal	907	989	Fatty	2^5	2^4	2^3	2^2
1-Octen-3-one	979	1070	Mushroom-like, earthy	2^6	2^7	2^4	2^4
Octanal	1003	1089	Soapy, citrus-like	2^7	2^6	2^2	2^2
(Z)-3-Hexenyl acetate	1008	1081	Fruity	2^5	2^4	2^3	1
Phenylacetaldehyde	1051	1174	Sweet, honey-like	2	2^2	1	2
(E)-2-Octenal	1060	1197	Fatty	2^3	2^3	1	<1
Guaiacol	1092	1232	Burnt	2^3	2^4	2^2	2^4
(Z)-3-Nonenal	1094	1184	Fatty	2^2	2	1	<1
Nonanal	1105	1192	Soapy, citrus-like	2^3	2^4	1	<1
2-Phenylethanol	1117	1278	Sweet, honey-like	2^3	2^4	2^3	2^3
Ethyl cyclohexyl-carboxylate	1133	1215	Fruity	<1	2	1	<1
(Z)-2-Nonenal	1147	1257	Fatty	2^5	2^7	1	2
(E,Z)-2,6-Nonadienal	1158	1276	Cucumber-like	2^4	2^5	2^3	2^4
(E)-2-Nonenal	1161	1275	Fatty, tallowy	2^4	2^4	1	2
2,4-Nonadienal	1194	1324	Deep-fried	1	2	<1	<1
(E,E)-2,4-Nonadienal	1216	1343	Deep-fried	2	2	<1	2
2-Decenal	1251	1356	Fatty	2^2	2^3	<1	2
(E)-2-Decenal	1263	1381	Fatty	2^3	2^3	2	2
(E,Z)-2,4-Decadienal	1294	1429	Deep-fried	2^3	2^4	<1	<1
(E,E)-2,4-Decadienal	1317	1452	Deep-fried	2^5	2^8	2^2	2^2
trans-4,5-Epoxy-(E)-2-decenal	1378	1551	Metallic	2^4	2^5	2^4	2^4
Vanillin	1421	1631	Vanilla-like	2	2	2	2^2

* All data are produced in the same laboratory

et al., 2004; Dhifi et al., 2005; Morales et al., 2005). In addition to these volatiles, linolenic, linoleic, and oleic autoxidation decomposition products, mainly aldehydes, and also aldehydes, alcohols, and esters deriving from biochemical transformations of amino acids such as isoleucine, leucin, phenylalanine, or valine (Schreier, 1984), have

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TABLE 4.6
Levels of some odorants in virgin olive oil samples differing in flavor

Compound	Concentration (μg/kg)							
	a	b	c	d	e	f	g	h
Hexanal	620	460	95	370	260	180	460	1510
(E)-2-Hexenal	1150	7400	730	1550	650	330	1640	170
(Z)-3-Hexenal	31	165	10	57	31	25	37	8
Octanal	130	125	100	140	85	70	220	210
(Z)-2-Nonenal	9	10	7	11	16	15	14	15
(E)-2-Nonenal	21	22	11	25	33	19	44	46
(E,E)-2,4-Decadienal	77	75	39	73	880	1330	245	640
1-Octen-3-one	2.7	4.1	0.9	1.9	1.5	1.7	5.7	13
Ethyl 2-methylpropanoate	3.5	2.4	2.7	2.3	2.2	1.0	62	115
Ethyl 2-methylbutanoate	1.8	2.9	1.9	2.4	4.0	0.7	48	105
Ethyl cyclohexylcarboxylate	4.9	12.3	7.7	5.5	3.8	3.2	3.0	2.7
(Z)-3-Hexenylacetate	9130	7880	130	13440	300	20	130	75
(Z)-3-Hexen-1-ol	470	1310	600	1140	165	65	720	570
2-Phenylethanol	6170	6620	5770	7010	810	17	320	1470
Acetic acid	2880	5740	4610	3560	3180	2390	4970	14340

a low odor threshold and can contribute to the aroma of olive oil.

A greater number of volatile compounds have been found in virgin olive oils of poorer quality. Some of these volatiles give rise to sensory defects when they are present at high levels. They are produced by over-ripening of the fruit, significant attack of the fruits by molds and bacteria, when they are stored for a long period prior to oil extraction, and also by advanced autoxidation of the unsaturated fatty acids due to adverse storage conditions (Angerosa, 2002). Acids, esters, alcohols, aldehydes and ketones are mainly responsible for the most frequent off-flavors developed in virgin olive oil (Gutiérrez et al., 1981; Angerosa et al., 1996; Morales et al., 2000; Morales et al., 2005). According to Morales et al. (2005), the main contributors to the so called "fusty" off-flavor, which is characteristic for oils obtained from olive fruits being in an advanced stage of fermentation, are mainly ethyl butanoate, and also propanoic and butanoic acids. Responsible for the "mustiness-humidity" virgin olive oil off-flavor is 1-octen-3-ol, and to a lesser extent 1-octen-3-one. This off-flavor is characteristic for oils obtained from fruits piled under humid conditions for several days, giving rise to the development of various kinds of fungi. The main volatiles responsible for the "rancid" off-flavor are aldehydes which are decomposition products of linolenic, linoleic, and oleic acid hydroperoxides. Finally, the main odorants which contribute to the "winey-vinegary" off-flavor are acetic acid, 3-methylbutan-1-ol, and ethyl acetate.

The presence of terpene alcohols (Flath et al., 1973), monoterpene and sesquiterpene hydrocarbons in the virgin olive oil (Gutiérrez et al., 1981; Lanzón et al., 1994; Bortolomeazzi et al., 2001; Vichi et al., 2003), as well as aromatic hydrocarbons is

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also interesting (Olias et al., 1980; Gutiérrez et al., 1981; Vichi et al., 2003). However, it is not clear whether all aromatic hydrocarbons are naturally present compounds or contaminants (Biedermann et al., 1995).

Other compounds identified in the volatile fraction are ethyl esters. Oils obtained from altered olive fruits or olive pomace have been found to contain high levels of ethyl palmitate, ethyl oleate and ethyl linoleate, while the levels of these esters in extra virgin olive oils are low (Pérez-Camino et al., 2002).

Other Minor Constituents

Some classes of minor constituents are present only in the crude oil. Filtration reduces the initial levels to a great extent whereas refining results in their removal.

Phospholipids. Experimental work for the identification of phospholipids in olive oil is rather limited. Phosphatidylcholine, phospatidylethanolamine, phosphatitylinositol, and phosphatidylserine were reported to be the main phospholipids present in olive oil (Alter and Gutfinger, 1982). The fatty acid composition was found to be similar to that of triacylglycerols.

In a recent report Boukhchina et al. (2004) identified glycerophospholipids present in olive oil by liquid chromatography-mass spectrometry (GC-MS). The LC effluent was directly introduced into the mass spectrometer through an electrospray capillary while information about the fatty acid composition of each phospholipids class was given by the tandem mass spectra obtained for negatively charged lipids. Phosphatidylethanolamine, phosphatidylcholine, phosphatidylinositol, phosphatidic acid, and also phosphatidylglycerol were the phospholipids identified and quantified. Phosphatidylserine was not detected. However, the study was only based on one sample from the retail market.

The level of phospholipids may be important because these compounds have an antioxidant activity. According to Pokorny and Korczak (2001), these lipids may act as synergists (regeneration of antioxidants such as α-tocopherol or other phenols) or as metal scavengers. At high levels, however, phospholipids may cause foaming or darkening during frying. The possible contribution of phospholipids to the oxidative stability of olive oil has not been studied. Koidis and Boskou (unpublished data) determined phosphorous in cloudy olive oils, filtered oils and refined oils. Values obtained were in the range of 1-6 mg P/kg oil (n=26), corresponding to approximately 20-156 mg phospholipids/kg oil. The higher level of phospholipids in the unfiltered oils may be an additional antioxidant factor to phenols. These veiled oils were found to be more stable to oxidation and this was attributed to the higher levels of polar phenols (Tsimidou et al., 2004).

Proteins. Mainly in unfiltered oils minute quantities of proteins may be detected (see cloudy olive oil, Chapter 11 Storage and Packing).

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References

Alter M., T. Gutfinger, Phospholipids in Several Vegetable Oils. *Riv. Ital. Sost. Grasse* 59: 14-18, (1982).

Andrikopoulos N., M. Hassapidou, A. Manoukas, The Tocopherol Content of Greek Olive Oils. *J. Sci. Food Agric*. 46: 503-509, (1989).

Angerosa F., C. Basti, The Volatile Composition of Samples From the Blend of Monovarietal Olive Oils and From the Processing of Mixtures of Olive Fruits. *Eur. J. Lipid Sci. Technol.* 105: 327-332, (2003).

Angerosa F., C. Basti, R. Vito, Virgin Olive Oil Volatile Compounds From Lipoxygenase Pathway and Characterization of Some Italian Cultivars. *J. Agric. Food Chem.* 47: 836-839, (1999).

Angerosa F., C. Basti, R. Vito, et al., Effect of Fruit Removal on the Production of Virgin Olive Oil Volatile Compounds. *Food Chem.* 67: 295-299, (1999)a.

Angerosa F., L. Camera, N. d' Alessandro, et al., Characterization of Seven New Hydrocarbon Compounds Present in the Aroma of Virgin Olive Oil. *J. Agric. Food Chem.* 46: 648-653, (1998).

Angerosa F., N. d' Alessandro, C. Basti, et al., Biogeneration of Volatile Compounds in Virgin Olive Oil: Their Evolution in Relation to Malaxation Time. *J. Agric. Food Chem.* 46: 2940-2944, (1998)a.

Angerosa F., B. Lanza, V. Marsilio, Biogenesis of Fusty Defect in Virgin Olive Oils. *Grasas Aceites* 47: 142-150, (1996).

Angerosa F., R. Mostallino, C. Basti, et al., Influence of Malaxation Temperature and Time on the Quality of Virgin Olive Oil. *Food Chem.* 72: 19-28, (2001).

Angerosa F., R. Mostallino, C. Basti, et al., Virgin Olive Oil Odor Notes: Their Relationships With Volatile Compounds From the Lipoxygenase Pathway and Secoiridoid Compounds. *Food Chem.* 68: 283-287, (2000).

Angerosa F., Influence of Volatile Compounds on Virgin Olive Oil Quality Evaluated by Analytical Approaches and Sensor Panels. *Eur. J. Lipid Sci. Technol.* 104: 639-660, (2002).

Aparicio R., G. Luna, Characterisation of Monovarietal Virgin Olive Oils. *Eur. J. Lipid Sci. Technol.* 104: 614-627, (2002).

Aparicio R., M. Morales, Characterization of Olive Ripeness by Green Aroma Compounds of Virgin Olive Oil. *J. Agric. Food Chem.* 46: 1116-1122, (1998).

Belitz H.-D., W. Grosch, P. Schieberle (eds), *Food Chemistry (3rd edition)*. Berlin: Springer Verlag, (2004).

Beltran G., A. Aguilera, C. del Rio, et al., Influence of Fruit Ripening Process on the Natural Antioxidant Content of Hojiblanca Virgin Olive Oils. *Food Chem.* 89: 207-215, (2005).

Benincasa C., A. De Nino, N. Lombardo, et al., Assay of Aroma Active Components of Virgin Olive Oils from Southern Italian Regions by SPME-GC/Ion Trap Mass Spectrometry. *J. Agric. Food Chem.* 51: 733-741, (2003).

Bianchi G., C. Murelli, G. Vlahov, Surface Waxes From Olive Fruits. *Phytochemistry* 31: 3503-3506, (1992).

Copyright © 2006 by AOCS Press

Bianchi G., N. Pozzi, G. Vlahov, Pentacyclic Triterpene Acids in Olives. *Phytochemistry* 37: 205-207, (1994).

Biedermann M., K. Grob, G. Morchio, On the Origin of Benzene, Toluene, Ethylbenzene and Xylene in Extra Virgin Olive Oil. *Z. Lebensm. Unters. Forsch.* 200: 266-272, (1995).

Blekas G., H. Guth, W. Grosch, In. H. Maarse, D. G. Van der Heij (eds). *Trends in Flavor Research*. Amsterdam: Elsevier Sciences, pp 499-502, (1994).

Blekas G., H. Guth, In. G. Charalambous (ed), *Food Flavors: Generation, Analysis and Process Influence.* Amsterdam: Elsevier Sciences, pp 419-427, (1995).

Bortolomeazzi R., P. Berno, L. Pizzale, et al., Sesquiterpene, Alkene, and Alkane Hydrocarbons in Virgin Olive Oils of Different Varieties and Geographical Origins. *J. Agric. Food Chem.* 49: 3278-3283, (2001).

Boskou D., I. Morton, Changes in the Sterol Composition of Olive Oil on Heating. *J. Sci. Food Agric.* 26: 1149-1153, (1975).

Boskou D., G. Stefanou, M. Konstandinidis, Tetracosanol and Hexacosanol Content of Greek Olive Oils. *Grasas Aceites* 34: 402-404, (1983).

Boskou D., I. Vlachopoulou, On the Level of Steryl Esters in Olive Oil. *Lebensm.-Wiss. u. Technol.* 19: 156-157, (1986).

Boskou D., In Boskou D (ed), *Olive Oil: Chemistry and Technology*. Champaign: AOCS Press, pp 101-120, (1996).

Boukhchina S., K. Sebai, A. Cherif, et al., Identification of Glycerophospholipids in Rapeseed, Olive, Almond and Sunflower Oils by LC-MS and LC-MS-MS. *Can. J. Chem.* 82: 1210-1215, (2004).

Calapaj R., S. Chiricosta, G. Saija, et al., Evaluation of Gas Chromatographic and Spectrophotometric Analytical Results to Check the Presence of Seed Oils in Olive Oil Samples. *Riv. Ital. Sost. Grasse* 70: 585-594, (1993).

Camera L., F. Angerosa, The Higher Alcohols of Olive Oil. Their Development With the Progressive Ripening of the Olives. *Riv. Ital. Sost. Grasse* 55: 138-146, (1978).

Camera L., F. Angerosa, A. Cucurachi, The Influence of Olives Storage on the Constituents of Oil Sterolic Fraction. *Riv. Ital. Sost. Grasse* 55: 107-112, (1978).

Caputo R., L. Mangoni, P. Monaco, et al., Triterpenes in Husks of *Olea europaea*. *Phytochemistry* 13: 1551-1552, (1974).

Cavalli J.-F., X. Fernandez, L. Lizzani-Cuvelier, et al., Characterization of Volatile Compounds of French and Spanish Virgin Olive Oils by HS-SPME: Identification of Quality-freshness Markers. *Food Chem.* 88: 151-157, (2004).

Cavalli J.-F., X. Fernandez, L. Lizzani-Cuvelier, et al., Comparison of Static Headspace, Headspace Solid-phase Microextraction, Headspace Sorptive Extraction, and Direct Thermal Desorption Techniques on Chemical Composition of French Olive Oils. *J. Agric. Food Chem.* 51: 7709-7716, (2003).

Cert A., J. Alba, M. León-Camacho, et al., Effects of Talk Addition and Operating Mode on the Quality and Oxidative Stability of Virgin Olive Oils Obtained by Centrifugation. *J. Agric. Food Chem*. 44: 3930-3934, (1996).

Copyright © 2006 by AOCS Press

Cert A., J. Alba, C. Pérez-Camino, et al., Influence of Extraction Methods on the Characteristics and Minor Components of Extra Virgin Olive Oil. *Olivae* 79: 41-50, (1999).

Chryssafidis D., P. Magos, V. Kiosseoglou, et al., Composition of Total and Esterified 4α-monomethylsterols and Triterpene Alcohols in Virgin Olive Oil. *J. Sci. Food Agric.* 58: 581-583, (1992).

Codex Alimentarius Standard for Olive Oils and Olive Pomace Oils, CODEX STAN 33-33-1981 (Rev.2-2003)

Commission of the European Communities. Regulation No 2568/91. *Official Journal of the European Communities* No L 248, (1991).

Commission of the European Communities. Regulation No 282/98. *Official Journal of European Communities* No L 285, (1998).

Commission of the European Communities. Regulation No 1989/03. *Official Journal of the European Communities* No L 295, (2003).

Conte L., M. Caboni, G. Lercker, Olive Oils Produced in Romagne. Note 1. Oils From Lamone River Valley. *Riv. Ital. Sost. Grasse* 70: 157-160, (1993).

De Blas O., A. del Valle González, Determination of Sterols by Capillary Column Gas Chromatography. Differentiation Among Different Types of Olive Oil: Virgin, Refined and Solvent-extracted. *JAOCS* 73: 1685-1689, (1996).

De Leonardis A., V. Macciola, M. De Felice, Rapid Determination of Squalene in Virgin Olive Oils Using Gas-liquid Chromatography. *It. J. Food Sci*. 1: 75-80, (1998).

Dhifi W., F. Angerosa, A. Serraiocco, et al., Virgin Olive Oil Aroma: Characterization of Some Tunisian Cultivars. *Food Chem.* 93: 697-701, (2005).

Esti M., L. Cinquanta, A. Carrone, et al., Anti-oxidative Compounds and Quality Parameters in Virgin Olive Oils Produced in Molise. *Riv. Ital. Sost. Grasse* 73: 147-150, (1996).

Fedeli E., Olive Oil Technology. *Olivae* 45: 20-23, (1993).

Fedeli E., N. Cortesi, Quality, Origin and Technology of Virgin Olive Oils. *Riv. Ital. Sost. Grasse* 70: 419-426, (1993).

Flath R., R. Forrey, D. Guadagni, Aroma Components of Olive Oil. *J. Agric. Food Chem.* 21: 948-952, (1973).

Frega N., F. Bocci, G. Lercker, Direct Gas Chromatographic Analysis of the Unsaponifiable Fraction of Different Oils With a Polar Capillary Column. *JAOCS* 69: 447-455, (1992).

Frega N., F. Bocci, G. Lercker, Free Fatty Acids and Diacylglycerols as Quality Parameters for Extra Virgin Olive Oil. *Riv. Ital. Sost. Grasse* 70: 153-156, (1993).

Gallardo-Guerrero L., B. Gandul-Rojas, M. Roca, et al., Effect of Storage on the Original Pigment Profile of Spanish Virgin Olive Oil, *JAOCS* 82: 33-39, (2005).

Gandul-Rojas B., M. Mínguez-Mosquera, Chlorophyll and Carotenoid Composition in Virgin Olive Oils From Various Spanish Olive Varieties, *J. Sci. Food Agric.* 72: 31-39, (1996).

Grams G., K. Eskins, Dye-sensitized Photoxidation of Tocopherols: Correlation Between Singlet Oxygen Reactivity and Vitamin E Activity. *Biochemistry* 11: 606-608, (1972).

Grob K., M. Lanfranchi, C. Mariani, Evaluation of Olive Oils Through the Fatty Alcohols, the

Copyright © 2006 by AOCS Press

Sterols and Their Esters by Coupled LC-GC. *JAOCS* 67: 626-634, (1990).

Grosch W., Determination of Potent Odorants in Foods by Aroma Extract Dilution Analysis (AEDA) and Calculation of Odor Activity Values (OAVs). *Flavor and Fragrance Journal* 9: 147-158, (1994).

Gunstone F., J. Harwood, F. Padley (eds), *The Lipid Handbook (2nd Edition)*. London: Chapman and Hall, (1994).

Guth H., W. Grosch, A Comparative Study of the Potent Odorants of Different Virgin Olive Oils. *Fat Sci. Technol.* 93: 335-339, (1991).

Guth H., W. Grosch, Quantitation of Potent Odorants of Virgin Olive Oil by Stable-isotope Dilution Assays. *JAOCS* 70: 513-518, (1993).

Gutiérrez R., C. Dobarganes, F. Gutiérrez, et al., Volatile Components in the Aroma of Virgin Olive Oil. V. Oils Obtained From "Atrojado" Fruits. *Grasas Aceites* 32: 299-303, (1981).

Gutiérrez F., B. Jímenez, A. Ruiz, et al., Effect of Olive Ripeness on the Oxidative Stability of Virgin Olive Oil Extracted From the Varieties Picual and Hojiblanca and on the Different Components Involved. *J. Agric. Food Chem.* 47: 121-127, (1999).

Gutiérrez F., I. Varona, M. Albi, Relation of Acidity and Sensory Quality With Sterol Content of Olive Oil From Stored Fruits. *J. Agric. Food Chem.* 48: 1106-1110, (2000).

Homberg E., B. Bielefeld, Freie und Gebundene Sterine in Pflanzenfetten. *Fette-Seifen-Anstrichmittel* 87: 61-66, (1985).

International Olive oil Council (2003) "Trade standards applying to olive oil and olive pomace oil". COI/T.15/NC, no 3/Rev.1./5.December 2003.International Olive Oil Council, Madrid, Spain.

Itoh T., T. Tamura, T. Matsumoto, Sterol Composition of 19 Vegetable Oils. *JAOCS* 50: 122-125, (1973)a.

Itoh T., T. Tamura, T. Matsumoto, Methylsterol Composition of 19 Vegetable Oils. *JAOCS* 50: 300-303, (1973)b.

Itoh T., K. Yoshida, T. Yatsu, et al., Triterpene Alcohols and Sterols of Spanish Olive Oil. *JAOCS* 58: 545-550, (1981).

Kamal-Eldin A., L. Appelqvist, The Chemistry and Antioxidant Properties of Tocopherols and Tocotrienols. *Lipids* 31: 671-701, (1996).

Kiosseoglou V., I. Vlachopoulou, D. Boskou, Esterified 4-monomethyl- and 4,4-dimethyl-sterols in Some Vegetable Oils. *Grasas Aceites* 38: 102-103, (1987).

Kiosseoglou V., P. Kouzounas, The Role of Diglycerides, Monoglycerides and Free Fatty Acids in Olive Oil Minor Surface-active Lipid Interaction With Proteins at Oil-water Interface. *J. Disp. Sci. Techn.* 14: 527-531, (1993).

Kornfeld A., 4-Demethyl, 4-monomethyl, and 4,4-dimethylsterols in Some Vegetable oils. *Lipids* 16: 306-314, (1981).

Koutsaftakis A., F. Kotsifaki, E. Stefanoudaki E, Effect of Extraction System, Stage of Ripeness, and Kneading Temperature on the Sterol Composition of Virgin Olive Oils. *JAOCS* 76: 1477-1481, (1999).

Copyright © 2006 by AOCS Press

Langmeier M., S. Ginsburg, P. Matile, Chlorophyll Breakdown in Senescent Leaves: Demonstration of Mg-decheletase Activity. *Physiol. Plant* 189: 347-353, (1993).

Lanuzza F., G. Micali, G. Calabro, Sterol Analysis in Olive Oil by Transesterification and HPLC-HRGC Coupling. *Riv. Ital. Sost. Grasse* 72: 105-109, (1995).

Lanzón A., T. Albi, A. Cert, et al., The Hydrocarbon Fraction of Virgin Olive Oil and Changes Resulting From Refining. *JAOCS* 71: 285-291, (1994).

Lanzón A., T. Albi, A. Guinda, Formation of a Δ^7 triterpene Alcohol in Refined Olive Oils. *JAOCS* 76: 1421-1423, (1999).

Leone A., M. Santoro, V. Liuzzi, et al., The Structure of Diglycerides and Their Occurrence in Olive Oils as a Means to Characterize High Quality Products *Riv. Ital. Sost. Grasse* 65: 613-622, (1988).

Lercker G., N. Frega, F. Bocci, et al., Volatile Constituents and Oxidative Stability of Virgin Olive Oils: Influence of the Kneading of Olive-paste. *Grasas Aceites* 50: 26-29, (1999).

Luaces P., A. Pérez, C. Sanz C., Role of Olive Seed in the Biogenesis of Virgin Olive Oil Aroma. *J. Agric. Food Chem.* 51: 4741-4745, (2003).

Mailer R. J., Variation in Oil Quality and Fatty Acid Composition in Australian Olive Oil. *Australian Journal of Experimental Agriculture* 45: 115-119, (2005).

Manzi P., G. Panfili, M. Esti, et al., Natural Antioxidants in the Unsaponifiable Fraction of Virgin Olive Oils From Different Cultivars. *J. Sci. Food Agric*. 77: 115-120, (1998).

Mariani C., E. Fedeli, G. Morchio, Absolute Erythrodiol Content as a Possibility to Detect Olive Husk Oil in Olive Oil. *Riv. Ital. Sost. Grasse* 64: 359-363, (1987).

Mariani C., E. Fedeli, K. Grob, Evaluation of Free and Esterified Minor Components in Fatty Material. *Riv. Ital. Sost. Grasse* 68: 233-242, (1991).

Mariani C., S. Venturini, P. Bondioli, et al., Evaluation of the Variations Produced by Bleaching Process on More Meaningful Minor Components Free and Esterified in Olive Oil. *Riv. Ital. Sost. Grasse* 69: 393-399, (1992).

Mariani C., G. Bellan, K. Grob, On the Complexity of Sterol Fraction in Edible Fats and Oils Separation of "campesterol" Into Two epimers. *Riv. Ital. Sost. Grasse* 72: 97-104, (1995).

Mariani C., Ergosterol in Olive Oils. *Riv. Ital. Sost. Grasse* 75: 3-10, (1998).

Mariani C., G. Bellan, S. Venturini, et al., The Presence of Esterified Erythrodiol in Olive Oils. *Riv. Ital. Sost. Grasse* 75: 279-286, (1998).

Mínguez-Mosquera M., B. Gandul-Rojas, J. Garrido-Fernández, et al., Pigments Present in Virgin Olive Oil. *JAOCS* 67: 192-196, (1990).

Mínguez-Mosquera M., L. Rejano-Navarro, B. Gandul-Rojas, et al., Color-pigment Correlation in Virgin Olive Oil. *JAOCS* 68: 332-336, (1991).

Mínguez-Mosquera M., B. Gandul-Rojas, J. Garrido-Fernandez, Preparation of Cu(II) Complexes of Oxidized Chlorophylls and Their Determination by Thin-layer and High-performance Liquid Chromatography. *J. Chromatogr. A* 731: 261-271, (1996).

Montedoro G., M. Bertuccioli, F. Anichini, In. G. Charalambous (ed)., *Flavor of Foods and Beverages*. St. Louis: Academic Press, pp 247-281, (1978).

Copyright © 2006 by AOCS Press

Morales M., F. Angerosa, R. Aparicio, Effect of the Extraction Conditions of Virgin Olive Oil on the Lipoxygenase Cascade: Chemical and Sensory Implications. *Grasas Aceites* 50: 114-121, (1999).

Morales M., R. Aparicio, J. Ríos, Dynamic Headspace Gas Chromatographic Method for Determining Volatiles in Virgin Olive Oil. *J. Chromatogr.* 668: 455-462, (1994).

Morales M., R. Aparicio, Effect of Extraction Conditions on Sensory Quality of Virgin Olive Oil. *JAOCS* 76: 295-300, (1999).

Morales M., G. Luna, R. Aparicio, Sensory and Chemical Evaluation of Winey-vinegary Defect in Virgin Olive Oils. *Eur. Food Res. Technol.* 211: 222-228, (2000).

Morales M., G. Luna, R. Aparicio, Comparative Study of Virgin Olive Oil Sensory Defects. *Food Chem.* 91: 293-301, (2005).

Morchio G., R. De Anreis, E. Fedeli, Investigations of Total Sterols Content in the Olive Oil and Their Variation During Refining Process. *Riv. Ital. Sost. Grasse* 64: 185-192, (1987).

Nenadis N., M. Tsimidou, Determination of Squalene in Olive Oil Using Fractional Crystallization for Sample Preparation. *JAOCS* 79, 257-259, (2002).

Nergiz C., K. Ünal, The Effect of Extraction Systems on Triterpene Alcohols and Squalene Content of Virgin Olive Oil. *Grasas Aceites* 41: 117-121, (1990).

Ninni V., A Statistical Approach to the Biosynthetic Route of the Fatty Acids in Olive Oil: Cross-sectional and Time Series Analyses. *J.Sci.Food Agric*. 79: 2113-2121, (1999).

Olías J., A. Del Barrio, R. Gutiérrez, Volatile Components in the Aroma of Virgin Olive Oil. I. *Grasas Aceites* 29: 107-112, (1977).

Olías J., F. Gutiérrez, C. Dobarganes, et al., Volatile Components in the Aroma of Virgin Olive Oil. IV. Their Evolution and Influence in the Aroma During the Fruit Ripening Process in the "Picual" and "Hojiblanca" Varieties. *Grasas Aceites* 31: 391-402, (1980).

Olías J., A. Pérez, J. Ríos, et al., Aroma of Virgin Olive Oil: Biogenesis of the "green" Odor Notes. *J. Agric. Food Chem.* 41: 2368-2373, (1993).

Paganuzzi V., E. Leoni, On the Composition of Iranian Olive Oil. *JAOCS* 56: 924-930, (1979).

Paganuzzi V., Influence of Origin and Conservation on the Sterolic Composition of Nontreated Olive Oils III. Pressed Oils From Greece. *Riv. Ital. Sost. Grasse* 62: 399-409, (1985).

Paganuzzi V., Monoglycerides in Vegetable Oils. Note I: Monoglycerides in Vegetable Oils. *Riv. Ital. Sost. Grasse* 64: 411-414, (1987).

Paganuzzi V., Monoglycerides in Vegetable Oils. Note IV: Raw Oils of Law Unsaturation, *Riv. Ital. Sost. Grasse* 76: 457-471, (1999).

Parcerisa J., C. Isidre, J. Boatella, et al., Analysis of Olive and Hazelnut Oil Mixtures by HPLC-atmospheric Pressure Chemical Ionization Mass Spectrometry of Tricylglycerols and Gas Liquid Chromatography of Non-saponifiable Compounds (tocopherols and sterols). *J. Chromatography A* 881: 149-158, (2000).

Pérez A., P. Luaces, J. Ríos, et al., Modification of Volatile Compound Profile of Virgin Olive Oil Due to Hot-water Treatment of Olive Fruit. *J. Agric. Food Chem.* 51: 6544-6549, (2003).

Pérez-Camino C., A. Cert, Quantitative Determination of Hydroxy Pentacyclic Triterpene Acids.

Copyright © 2006 by AOCS Press

J. Agric. Food Chem. 47: 1558-1562, (1999).

Pérez-Camino C., W. Moreda, A. Cert, Effects of Olive Fruit Quality and Oil Storage Practices on the Diacylglycerol Content of Virgin Olive Oils. *J. Agric. Food Chem.* 49: 699-704, (2001).

Pérez-Camino C., W. Moreda, R. Mateos, et al., Determination of Esters of Fatty Acids With Low Molecular Weight Alcohols in Olive Oils. *J. Agric. Food Chem.* 50: 4721-4725, (2002).

Perrin J., Minor Components and Natural Antioxidants of Olives and Olive Oils. *Rev. Franç. Corps Gras* 39: 25-32, (1992).

Pokorny J., J. Korczak, In J. Pokorny, N. Yanishlieva, M. Gordon (ed). *Antioxidants in food*. Boca Raton: CRC Press, pp 311-330, (2001).

Preininger M., W. Grosch, Evaluation of Key Odorants of the Neutral Volatiles of Emmentaler Cheese by the Calculation of Odor Activity Values. *Lebensm.-Wiss. u. Technol.* 27: 237-244, (1994).

Psomiadou E., M. Tsimidou, Simultaneous HPLC Determination of Tocopherols, Carotenoids and Chlorophylls for Monitoring Their Effect on Virgin Olive Oil Oxidation. *J. Agric. Food Chem*. 46: 5132-5138, (1998).

Psomiadou E., M. Tsimidou, Pigments in Greek Virgin Olive Oils: Occurrence and Levels, *J. Sci. Food Agric*. 41: 640-647, (2001).

Psomiadou E., M. Tsimidou, Stability of Virgin Olive Oil. 1. Autoxidation Studies, *J. Agric. Food. Chem.* 50: 716-721, (2002)a.

Psomiadou E., M. Tsimidou, Stability of Virgin Olive Oil. 2. Photo-oxidation Studies, *J. Agric. Food. Chem.* 50: 722-727, (2002)b.

Psomiadou E., M. Tsimidou, D. Boskou, α-Tocopherol Content of Greek Virgin Olive Oils. *J.Agric. Food Chem.* 48: 1770-1775, (2000).

Rabascall N. H., J. B. Riera, Variations of the Tocopherols and Tocotrienols Content in the Obtention, Refining and Hydrogenation Processes of Edible Oils. *Gracas Aceites* 38: 145-148, (1987).

Rahmani M., A. Csallany, Chlorophyll and β-carotene Pigments in Moroccan Virgin Olive Oils Measured by High Performance Liquid Chromatography, *JAOCS* 68: 672-674, (1991).

Ranalli A., Carotenoids in Virgin Olive Oil. Effect of Technology, *Ital. J. Food Sci.* 1: 53-57, (1992).

Ranalli A., F. Angerosa, Integral Centrifuges for Olive Oil Extraction. The Qualitative Characteristics of Products. *JAOCS* 73: 417-422, (1996).

Ranalli A., A. Serraiocco, Evaluation of Characteristics of Olive Oil Produced by Innovative or Traditional Processing Technologies. *Riv. Ital. Sost. Grasse* 73: 303-315, (1996).

Ranalli A., G. De Mattia, M. Ferrante, Comparative Evaluation of the Olive Oil Given by a New Processing System. *Int. J. Food Sci. Technol.* 32: 289-297, (1997).

Ranalli A., M. Ferrante, G. De Mattia, et al., Analytical Evaluation of Virgin Olive Oil of First and Second Extraction. *J. Agric. Food Chem.* 47: 417-424, (1999).

Ranalli A., G. Modesti, M. Patumi, et al., The Compositional Quality and Sensory Properties of Virgin Olive Oil From a New Olive Cultivar – I-77. *Food Chem.* 69: 37-46, (2000).

Copyright © 2006 by AOCS Press

Rao C., H. Newmark, B. Reddy, Chemopreventive Effect of Squalene on Colon Cancer. *Carcinogenesis* 19: 287-290, (1998).

Reiners J., W. Grosch, Odorants of Virgin Olive Oils With Different Flavor Profiles. *J. Agric. Food Chem.* 46: 2754-2763, (1998).

Reiter B., E. Lorbeer, Analysis of the Wax Ester Fraction of Olive Oil and Sunflower Oil by Gas Chromatography and Gas Chromatography-mass Spectrometry. *JAOCS* 78: 881-888, (2001).

Ridolfi M., S. Terenziani, M. Patumi, et al., Characterization of the Lipoxygenases in Some Olive Cultivars and Determination of Their Role in Volatile Compound Formation. *J. Agric. Food Chem.* 50: 836-839, (2002).

Rivera del Álamo R., G. Fregapane, F. Aranda, et al., Sterol Composition of Cornicabra Virgin Olive Oil: the Campesterol Content Exceeds the Upper Limit of 4% Established by EU Regulations. *Food Chem.* 84: 533-537, (2004).

Salas J., J. Sánchez, The Decrease of Virgin Olive Oil Flavor Produced by High Malaxation Temperatures is Due to Inactivation of Hydroperoxide Lyase. *J. Agric. Food Chem.* 47: 809-812, (1999).

Salch Y., M. Grove, H. Takamura, et al., Characterization of a C-5,13-cleaving Enzyme of 13(S)-hydroperoxide of Linolenic Acid by Soybean Seed. *Plant Physiol.* 108: 1211-1218, (1995).

Salvador M., F. Aranda, G. Fregapane, Chemical Composition of Commercial Cornicabra Virgin Olive Oils From 1995/96 and 1996/97 crops. *JAOCS* 75: 1305-1311, (1998).

Sanchez Saez J., M. Herce Garraleta, T. Balea Otero, Identifcation of Cinnamic Acid Ethyl Ester and 4-vinylphenol in Off-flavor Olive Oils. *Anal. Chim. Acta* 247: 295-297, (1991).

Santinelli F., P. Damiani, W. W. Christie, The Triacylglycerol Structure of Olive Oil Determined by Silver Ion High Performance Liquid Chromatography in Combination With Stereospecific Analysis. *JAOCS* 69: 552-556, (1992).

Scano P., M. Casu, A. Lai, et al., Recognition and Quantitation of Cis-vaccenic and Eicosenoic Fatty Acids in Olive Oils by C-13 Nuclear Magnetic Resonance Spectroscopy. *Lipids*, 34: 757-759, (1999).

Schreier P. (ed)., *Chromatographic Studies of Biogenesis of Plant Volatiles*. Heidelberg: Hűthig Verlag, (1984).

Schwartz S., T. Lorenzo, Chlorophylls in Foods, *CRC Crit. Rev. Food Sci. Nutr*. 29: 1-17, (1990).

Servili M., P. Piacquadio, G. De Stefano, et al., Influence of a New Crushing Technique on the Composition of the Volatile Compounds and Related Sensory Quality of Virgin Olive Oil. *Eur. J. Lipid Sci. Technol.* 104: 483-489, (2002).

Servili M., R. Selvaggini, A. Taticchi, et al., Volatile Compound and Phenolic Composition of Virgin Olive Oil: Optimization of Temperature and Time of Exposure of Olive Pastes to Air Contact During the Mechanical Extraction Process. *J. Agric. Food Chem.* 51: 7980-7988, (2003).

Shioi Y., N. Tomita, T. Tsuchiya, et al., Conversion of Chlorophyllide to Pheophorbide by Mg-dechelating Substance in Extracts of Chenopodium Album, *Plant Physiol. Biochem.* 34: 41-47, (1996).

Smith T., G. Yang, D. Seril, et al., Inhibition of 4-(methylnitrosamino)-1-(3-pyridyl)-1-butanone-

Copyright © 2006 by AOCS Press

induced Lung Tumorigenesis by Dietary Olive Oil and Squalene. *Carcinogenesis* 19: 703-706, (1998).

Spyros, A., A. Philippidis, P. Dais, Kinetics of Diglyceride Formation and Isomerization in Virgin Olive Oils by Employing 31P NMR Spectroscopy. Formulation of a Quantitative Measure to Assess Olive Oil Storage History. *J. Agric. Food Chem.* 52:157-164, (2004).

State Chemical Laboratory (ed). *Analytical characteristics of Greek virgin olive oils (1992-1993).* Athens, (1994).

Strocchi A., A. Savino, Structural Modifications of Triterpene Alcohols During Olive Oil Refining. *Riv. Ital. Sost. Grasse* 66: 597-604, (1989).

Tiscornia E., N. Fiorina, F. Evangelisti, Chemical Composition of Olive Oil and Variations Induced by Refining. *Riv. Ital. Sost. Grasse* 59: 519-555, (1982).

Tsimidou M., A. Georgiou, A. Koidis, et al., Loss of Stability of "veiled" (cloudy) Virgin Olive Oils in Storage. *Food Chem.* 93: 377-383, (2004).

Tura D., P. Prenzler, D. Bedgood, et al., Varietal and Processing Effects on the Volatile Profile of Australian Olive Oils. *Food Chem.* 84: 341-349, (2004).

Vichi S., A. Castellote, L. Pizzale, et al., Analysis of Virgin Olive Oil Volatile Compounds by Headspace Solid-phase Microextraction Coupled to Gas Chromatography With Mass Spectrometric and Flamme Ionization Detection. *J. Chromatogr.* 83: 19-33, (2003).

Vlahov G., 13C Nuclear Magnetic Resonance Spectroscopy to Check 1,3-random, 2-random pattern of fatty acid distribution in olive oil triacylglycerols, *Spectroscopy* 19: 109-117, (2005).

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